



AI시대, 차세대 클라우드의 새로운 비전과 기술

openKcloud DevDay 2025

AI반도체로 바꾸는 클라우드의 미래 : 도전과 기회

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최현화 박사





A Giant Leap For LLM

LLM can shift the way people access and use knowledge
any fields, any language and at any time



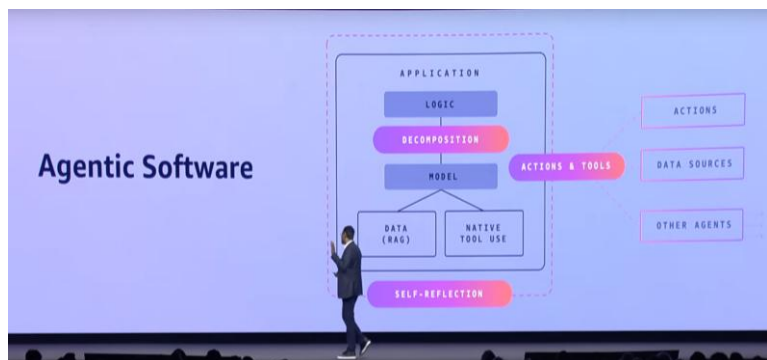
	2022~2023	2025
Gen AI	- GPT-3.5 - Claude - Google Bard - Llama 1 - Grok-0 - DeepSesk-coder/LLM - Phi-1	- OpenAI GPT-5 - Claude 4 - Gemini 2.5 pro - Llama 4 - Grok-4 - Deepseek-V3.1 - Phi-4
Capabilities	- Only text - Fair reasoning - Limited contextual understanding - Limited real-time data integration - Low personalization	- Multimodal(text, audio, images, video) - multistep problem-solving and nuanced analysis - Coherence in long dialogues - Real-time data integration(with tools) - Advanced personalization(user context)

How are cloud services evolving with the advancement of LLMs?

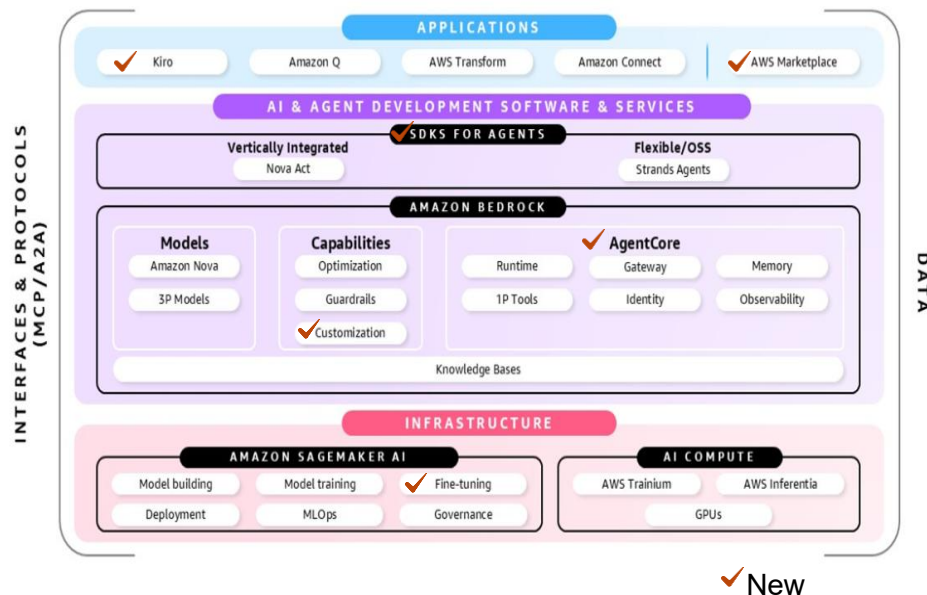
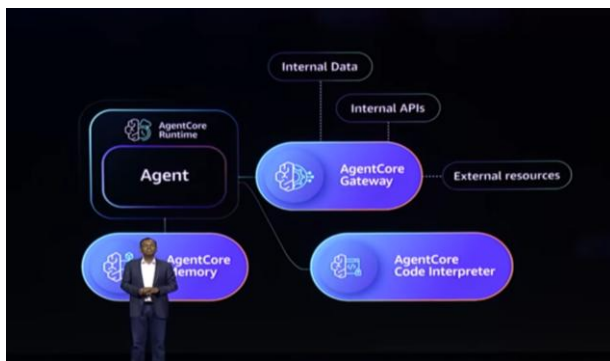


AWS: AI agents service as Software

- AI agent
 - Learn from their environment and adapt to changing conditions, enabling them to perform sophisticated workflows with accuracy
 - Proactive, Adaptable, Collaborative, Specialized



AWS Summit NY 2025

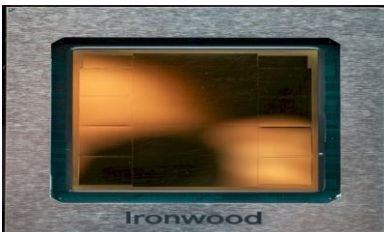




Google : The new way to cloud

• Automation AnyWhere

Infrastructure



- 10x Performance
- 42.5 EFLOPS
- 28x Power Efficiency

Model



- MoE Architecture
- 1 million Window
- Multimodal (video)
- Tool Use Intelligence

AI Platform



- Blackwell GPU (A4, A4X VMs)
- GKE Gateway, Pathway
- vLLM, Vertex AI

Multi-Agent



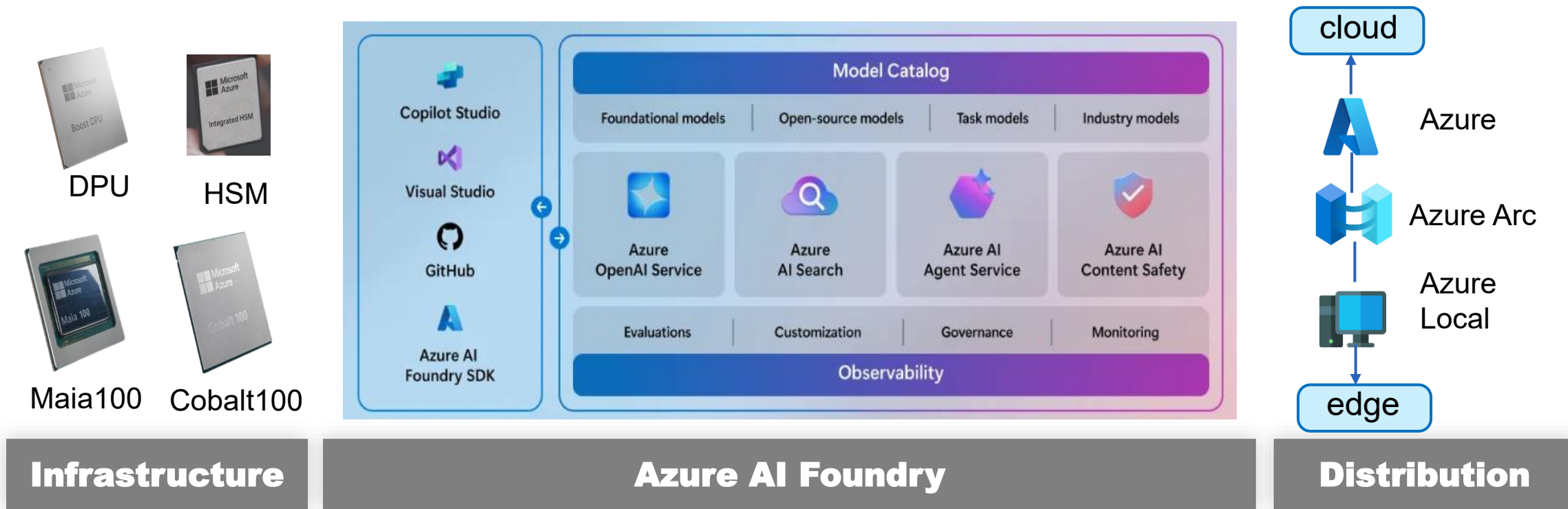
- Agent Development Kit
- Agent2Agent Protocol
- Agentspace, Vertex AI Agent Engine

Google Cloud Next 2025



Microsoft: Build agentic world

• Copilot with Agent service



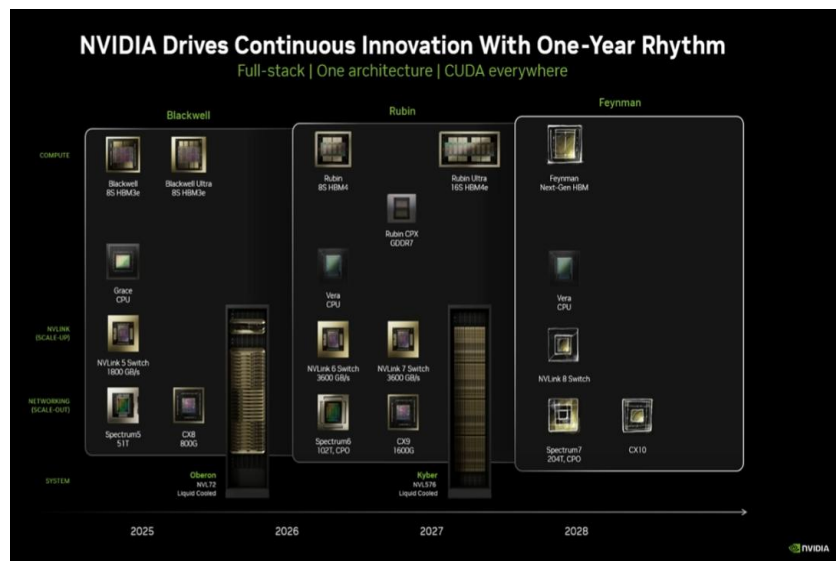


Next AI Cloud Platform

**Everybody is talking about Agentic AI.
What is the issue for next cloud platform?**



Problem1: Accelerator heterogeneity



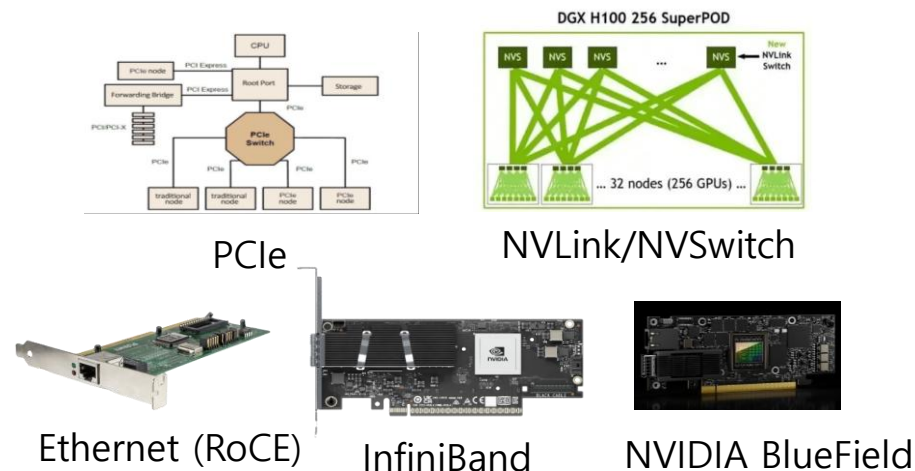
NVIDIA: a New GPU Every Year

NVIDIA GTC 2025

AI Accelerator (NPU)

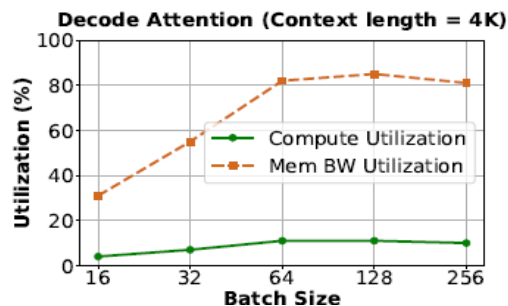
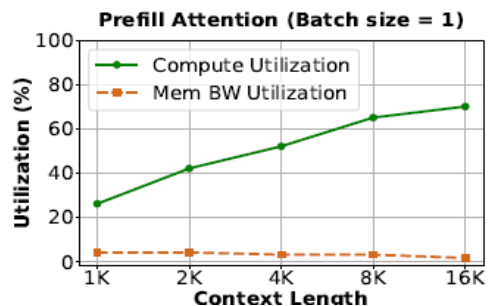
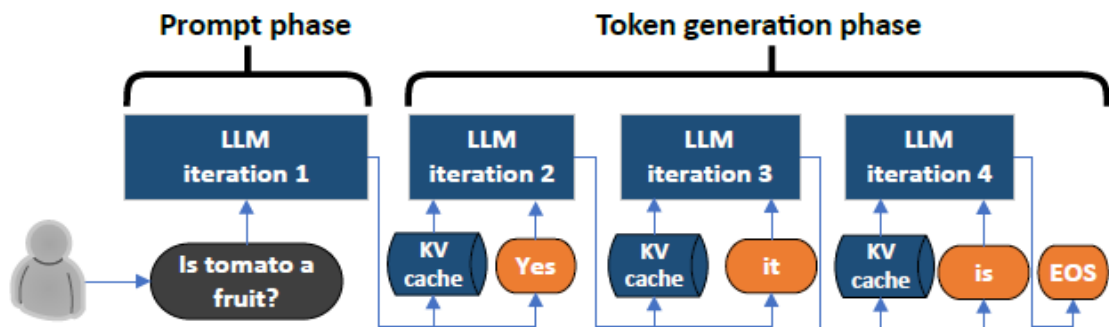


Inter/Intra Connect

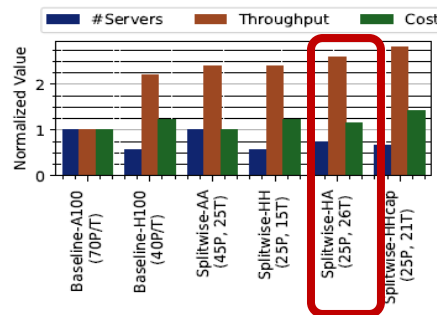




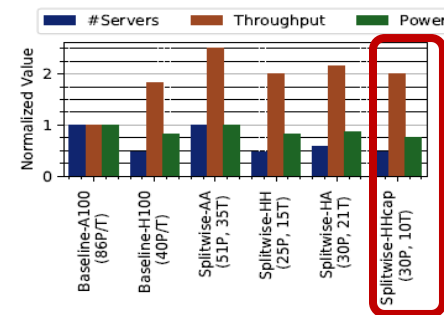
Challenge 1: No One-Size-Fits-All



POD-Attention(ASPLOS 2025)



(a) Iso-power.



(b) Iso-cost.

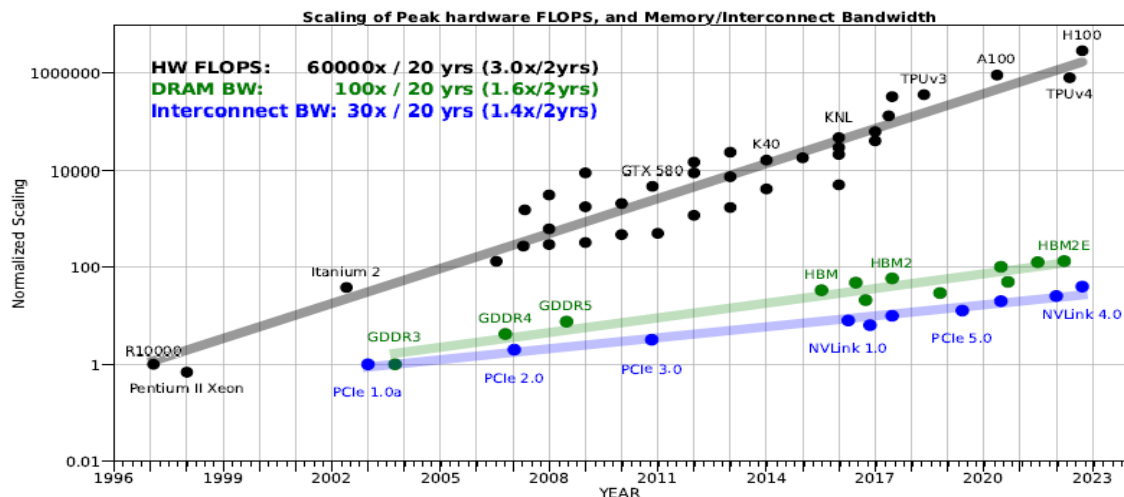
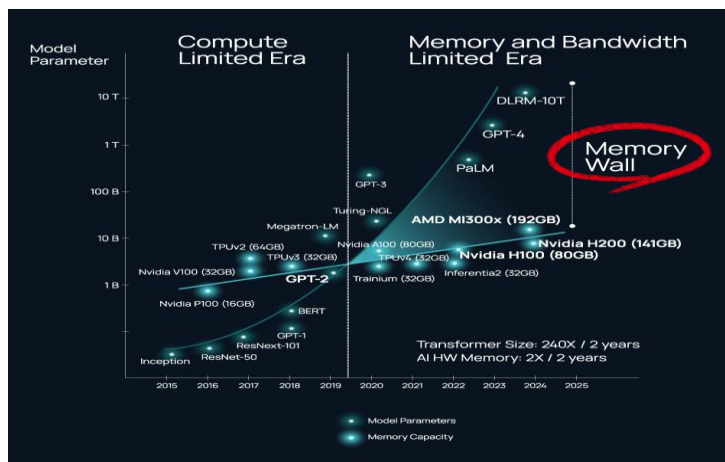
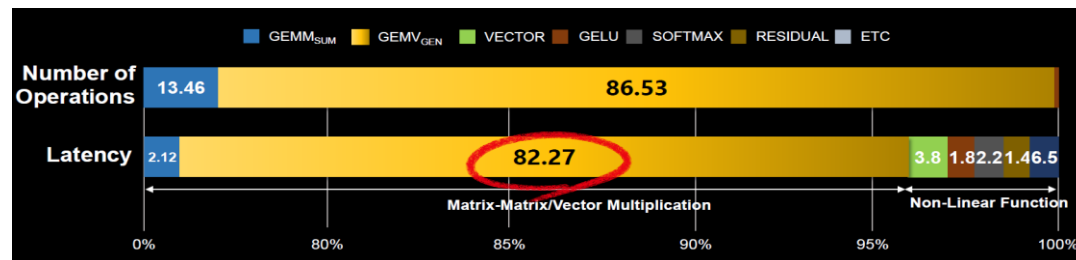
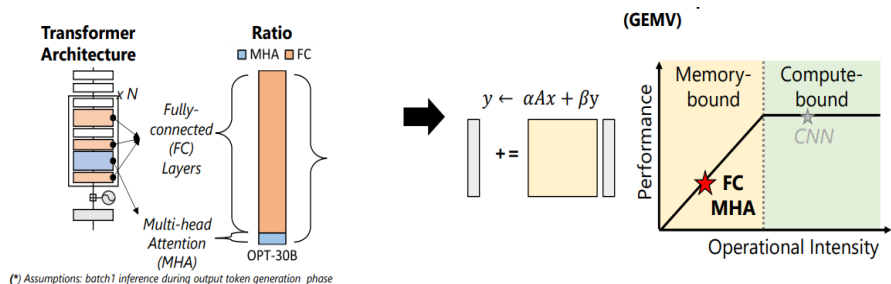
Fig. 18: Summary of throughput-optimized cluster designs.

Construct could with a heterogeneous mix of HW resources



Problem 2: Memory wall

LLM inference is a memory bound Job



<https://www.celestial.ai/technology-1>

AI and Memory Wall(Micro 2024)



Challenge 2: New memory device

PIM



SK hynix
AiMX



Samsung
HBM-PIM

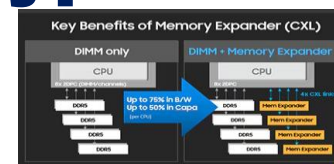
CXL Type3



Micron
CZ122



SK Hynix
CMM-DDR5



SMDK
(Scalable Memory Development Kit)



Astera Labs. Leo CXL
(Smart Memory Controller)

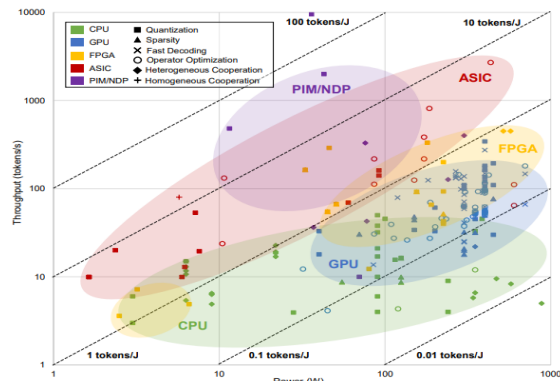
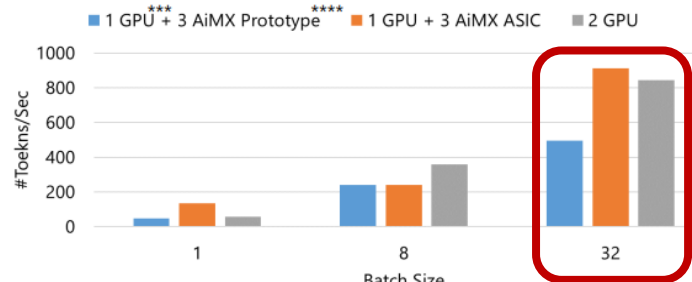


Figure 13: LLM (~7 billion parameters) decode stage throughput (batch size 1) vs power on different platforms with different optimization methods.

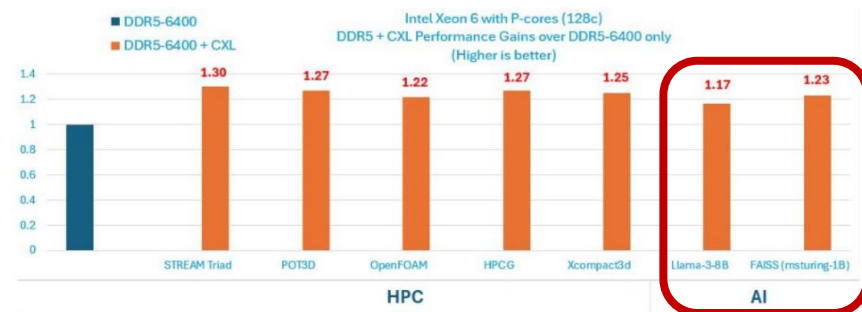
<https://dai.sjtu.edu.cn/project.html>

Performance Evaluation (OPT3-30B)



Note: FC layers in GPU, MHA in AiMX when batch size > 1
(***) H100 80GB GPU

SK hynix (Hot Chips 2024)

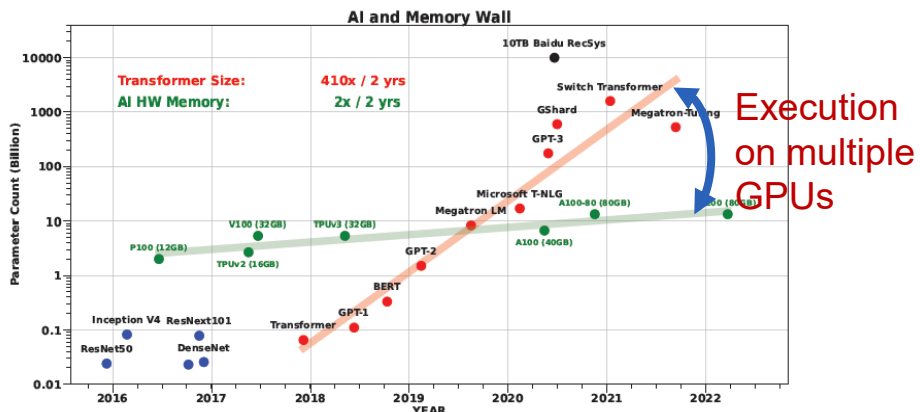


Micron (2024)

Boost LLM inference performance with next-gen memory



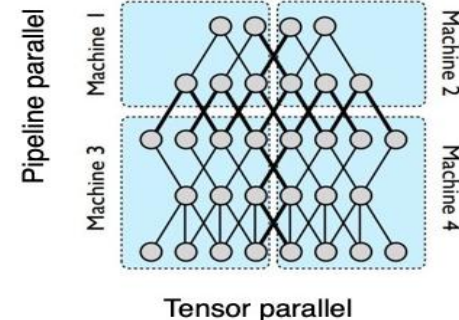
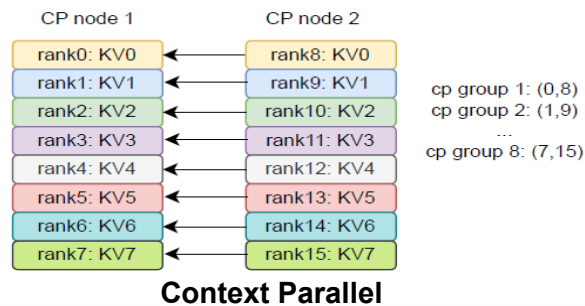
Problem 3: LLM Scaling



Chinchilla Scaling: 25.6 tokens/parameters



Parallelism



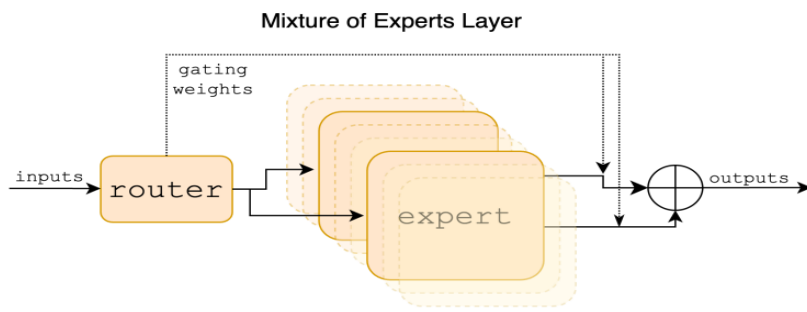
Quantization

Hardware	Weight-Only Quantization	Weight-Activation Quantization	KV Cache Quantization
CPU	✓	✗	✗
GPU	✓	✓	✓
FPGA	✗	✓	✗
ASIC	✗	✓	✗
PIM/NDP	✗	✓	✗

We want smarter LLM inference service.
That's the Problem



Challenge 3: MoE Architecture



	LLaMA 2 70B	GPT-3.5	Mixtral 8x7B
MMLU (MCQ in 57 subjects)	69.9%	70.0%	70.6%
HellaSwag (10-shot)	87.1%	85.5%	86.7%
ARC Challenge (25-shot)	85.1%	85.2%	85.8%
WinoGrande (5-shot)	83.2%	81.6%	81.2%
MBPP (pass@1)	49.8%	52.2%	60.7%
GSM-8K (5-shot)	53.6%	57.1%	58.4%
MT Bench (for Instruct Models)	6.86	8.32	8.30

Mixtral of Experts(2024)

Model Intelligence



<https://artificialanalysis.ai/models/llama-3-1-instruct-70b/providers>

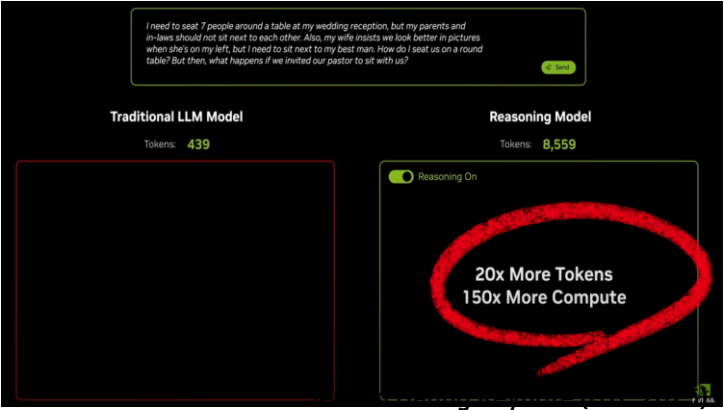
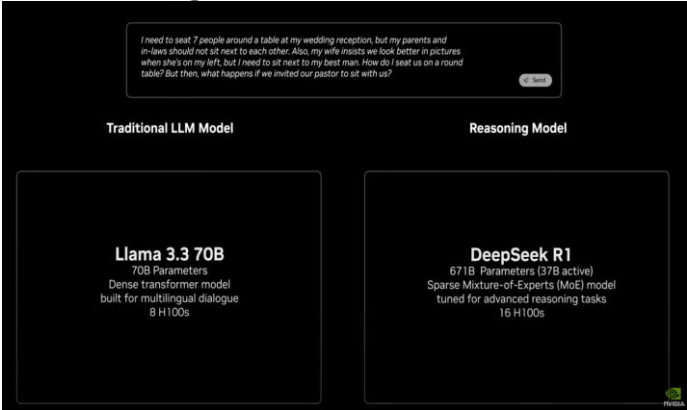
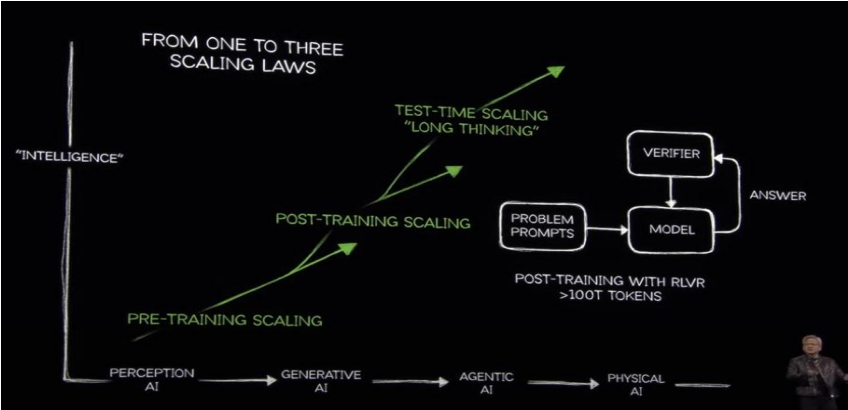
Provide more intelligent inference from multiple LLMs



Problem 4: LLM Inference Cost (Energy)

Knowledge AI => Thinking AI (with Long Context)

Thinking for 20 seconds



GPU	A100 (80GB)	H100 (80GB)	H200 (141GB)	B100 (180G)	B200 (180G)
Power	300W	500W	700W	700W ¹	1000W ²

¹Dell PowerEdge XE9680: HGB B100 700W
²Renovo HGX B200 1000W

Cloud- Daily Energy Consumption

- NVIDIA DGX B200(14.3kW) * 100
- 1 Day = 41.2MWh
14.3kW * 100 servers * 24 hours = 34,320kWh
PUE (1.2) = 34.3MWH * 1.2 = 41.2MWh

- Tesla model3: 7times around the Earth



Challenge 4: Scheduling

Configuration parameters	Perf	Temp	Power	Quality
Model Size (e.g., 70B→7B)	↑	↓	↓	↓↓
Quantization (e.g., FP16→FP8)	↑	↓	↓	↓
Parallelism (e.g., TP8→TP2)	↓	↑	↓	—
Frequency (e.g., 2GHz→1GHz)	↓	↓	↓	—
Batch Size (e.g., 64→16)	↓	↓	↓	—

Table 1. Impact on performance, temperature, power, and quality of an LLM inference server for multiple parameters.

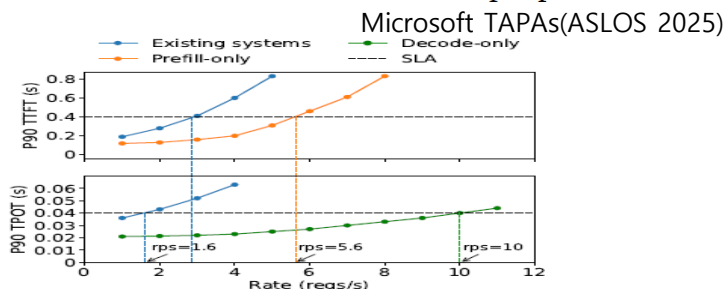


Figure 1: Performance when serving an LLM with 13B parameters under a synthetic workload with input length = 512 and output length = 64 on one NVIDIA 80GB A100. *Upper:* The P90 time-to-first-token (TTFT) latency comparing existing systems vs. a system serving only the prefill phase. *Down:* The P90 time-per-output-token (TPOT) latency comparing existing systems vs. a system serving only the decoding phase.

DistServe(OSDI 2025)

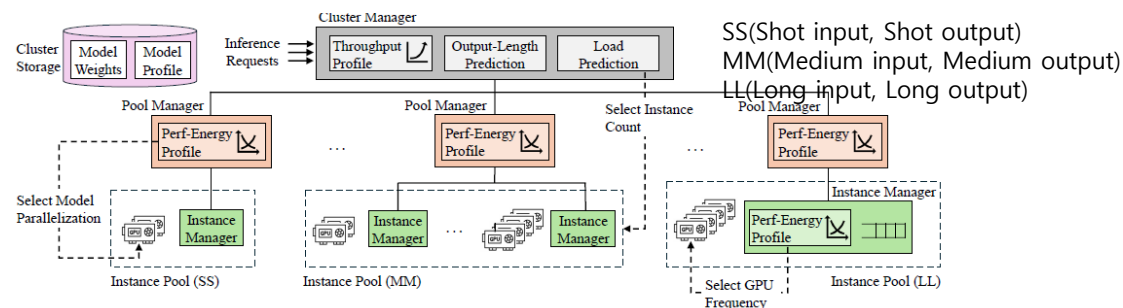


Fig. 4: DynamoLLM architecture: a hierarchy of controllers with cluster resources split into per request-type pools.

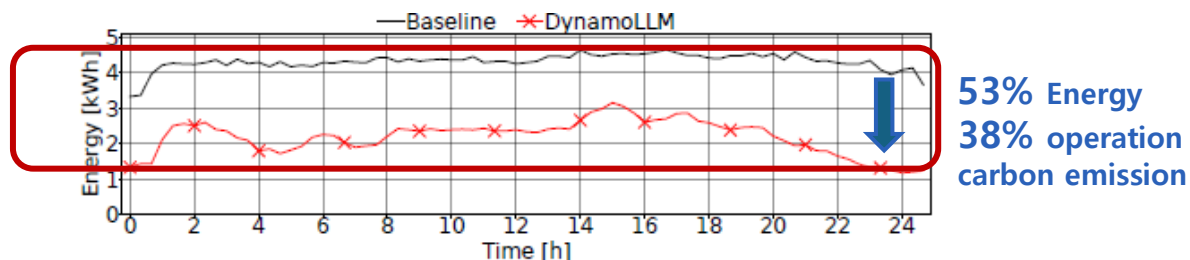


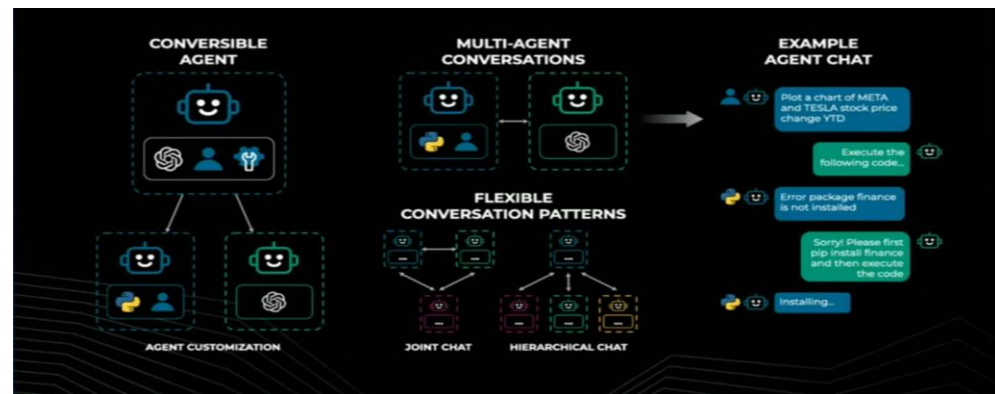
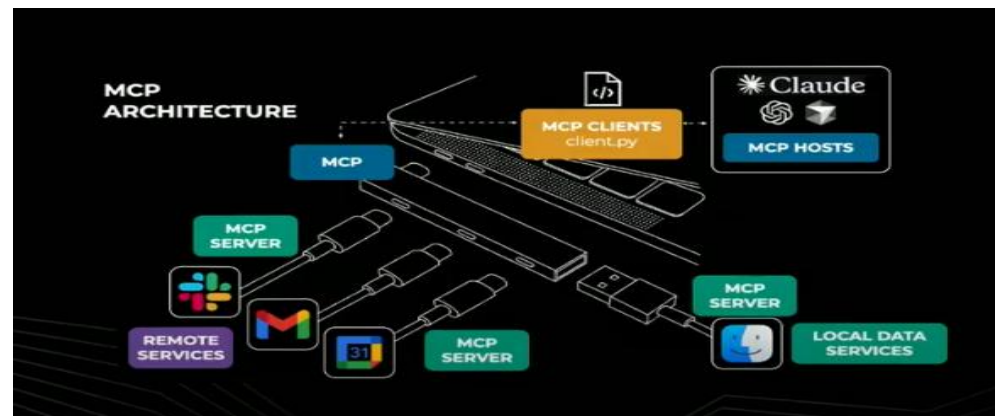
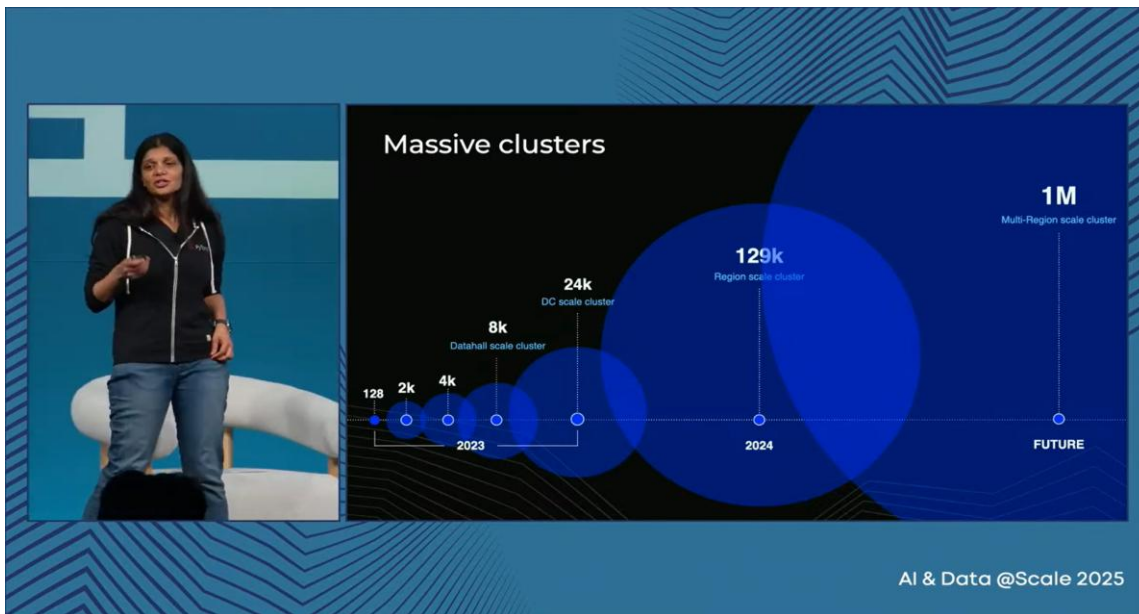
Fig. 15: Energy consumption of SinglePool and DynamoLLM with 1-day production traces running open-source Llama2-70B model [67].

DynamoLLM(OSDI 2025)

Reconfigure the LLM inference clusters to improve energy-efficiency



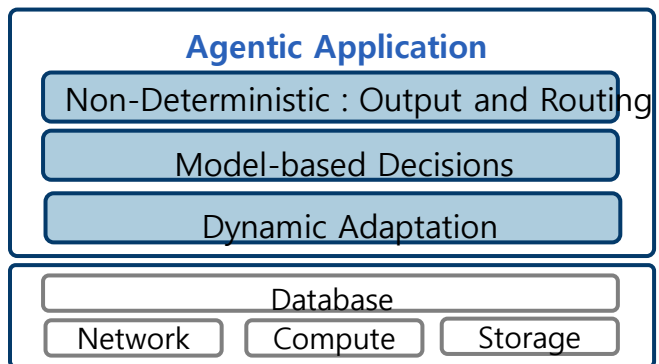
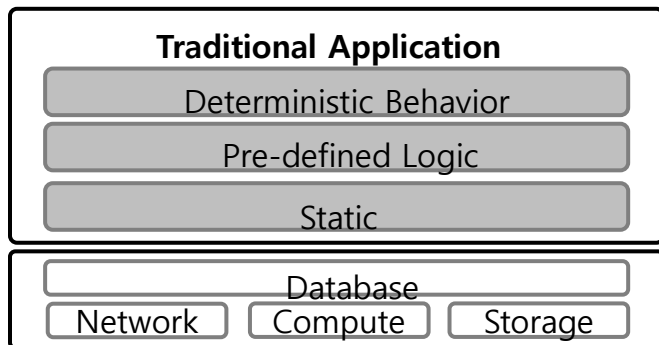
Problem 5: Massive Agents



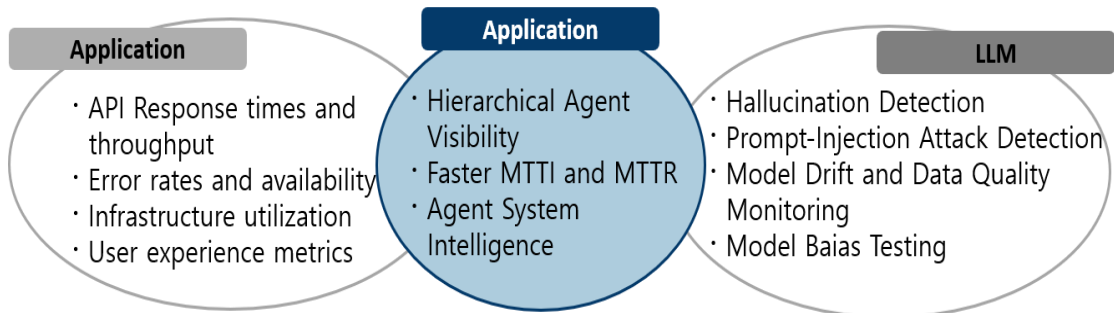
How to manage applications and services?



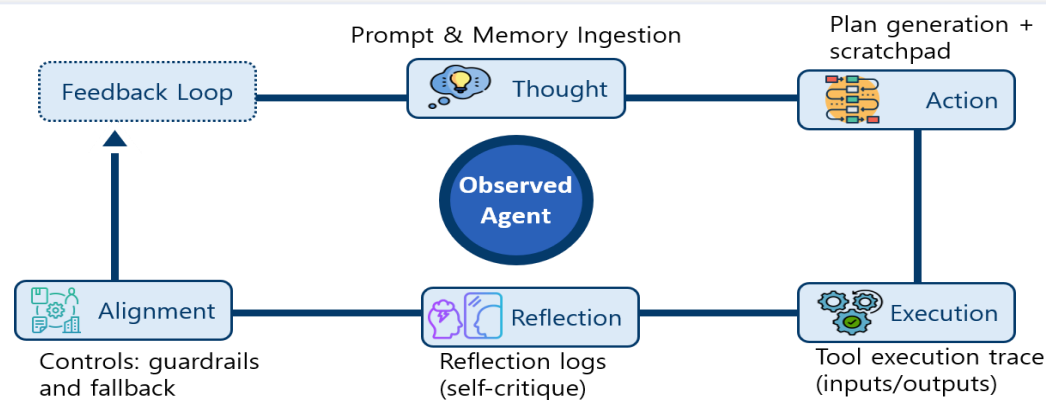
Challenge 5: Monitoring



Performance



Observed Agent



Provide agentic observability



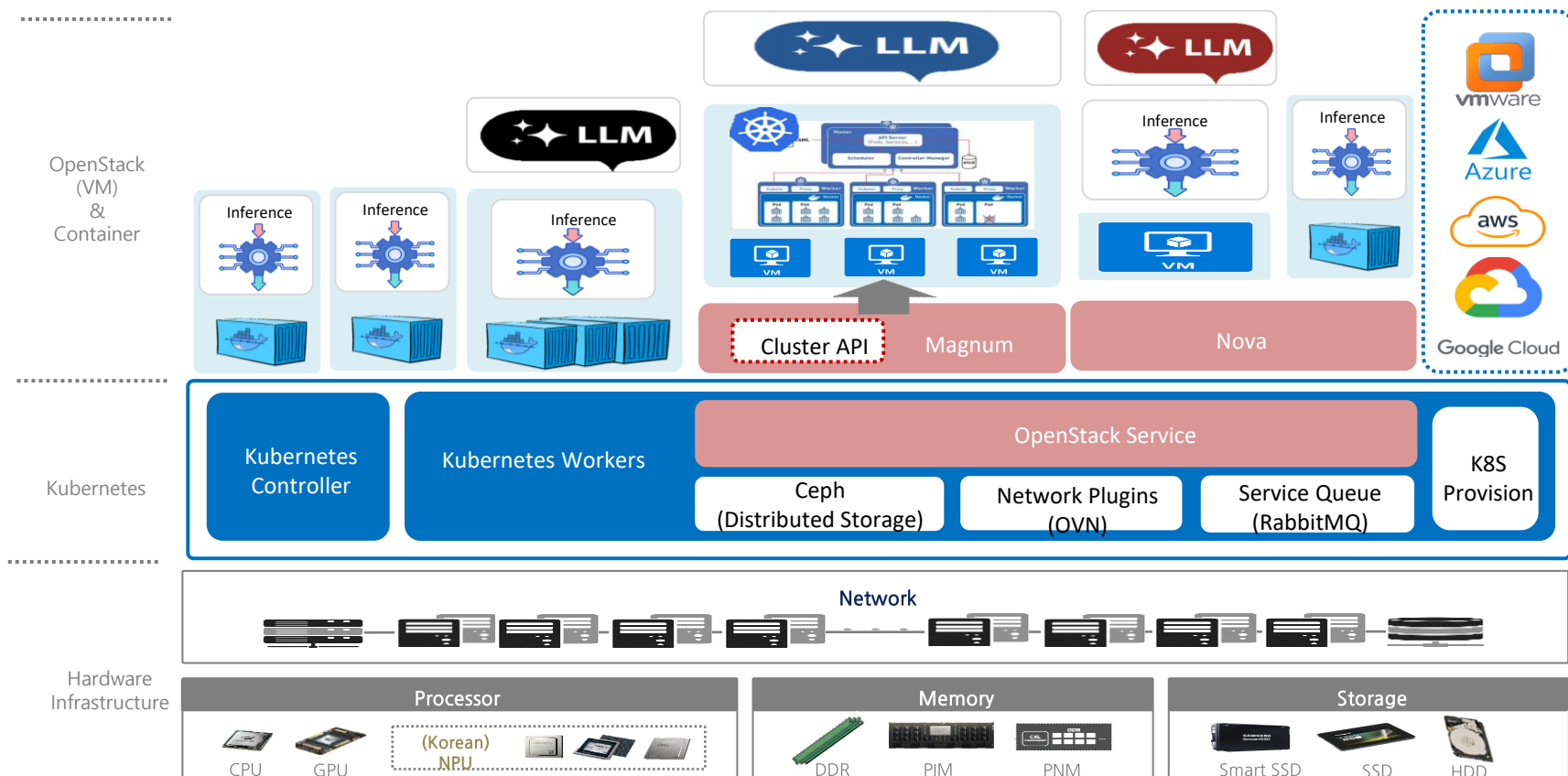
openKcloud

**We development openKcloud
to these solution.**



LLM Platform as a Services

Stable and Rapidly Scalable AI Semiconductor-based LLM service Clusters

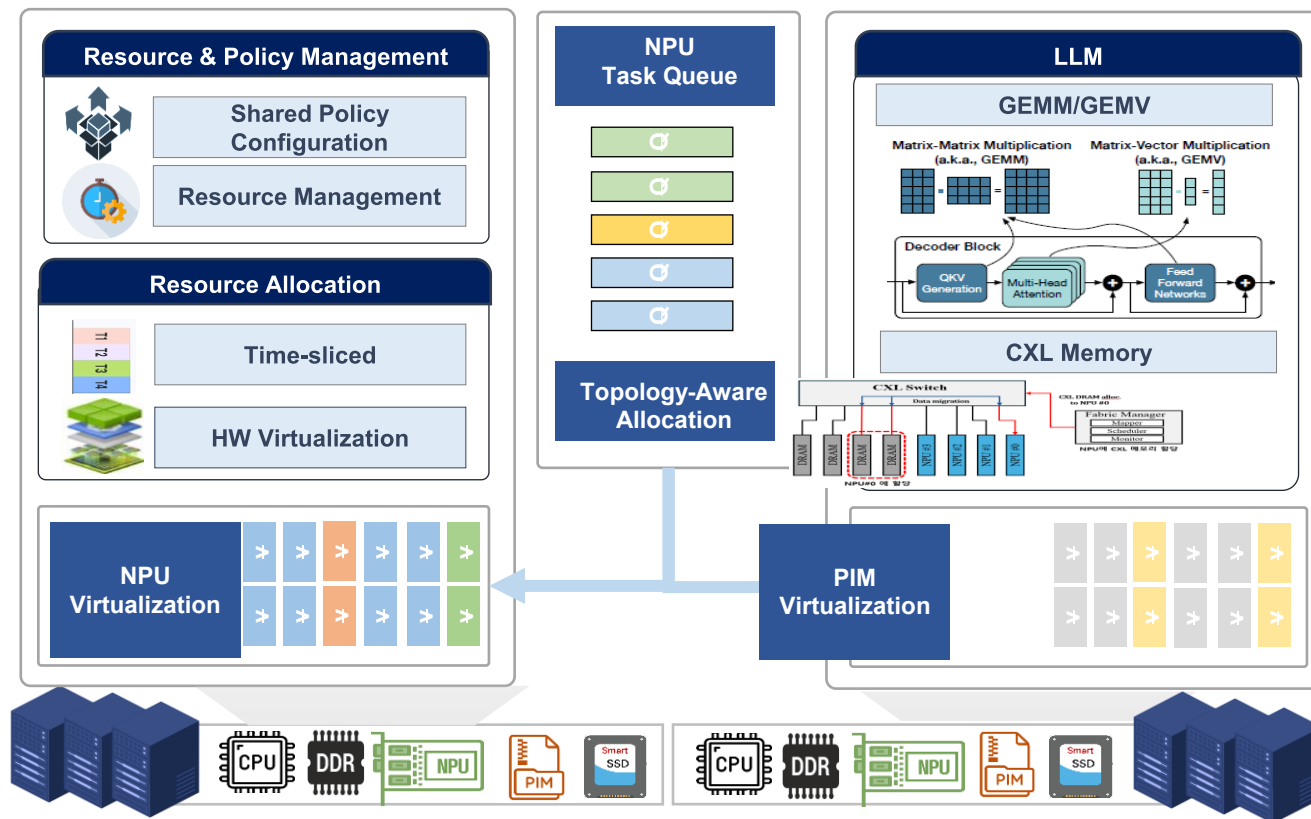




Heterogeneous AI semiconductors Management

Maximization of AI Semiconductor Utilization

- Breaking free from NVIDIA vendor lock-in with heterogeneous, homegrown AI chips
- AI-Semiconductor Sharing Across Multi-Clouds through Logical Partitioning & Time-Sliced Virtualization
- Compute and Storage Acceleration for LLM Cloud Services Leveraging PIM
- Topology-Aware Resource Co-Allocation to Minimize Data Movement

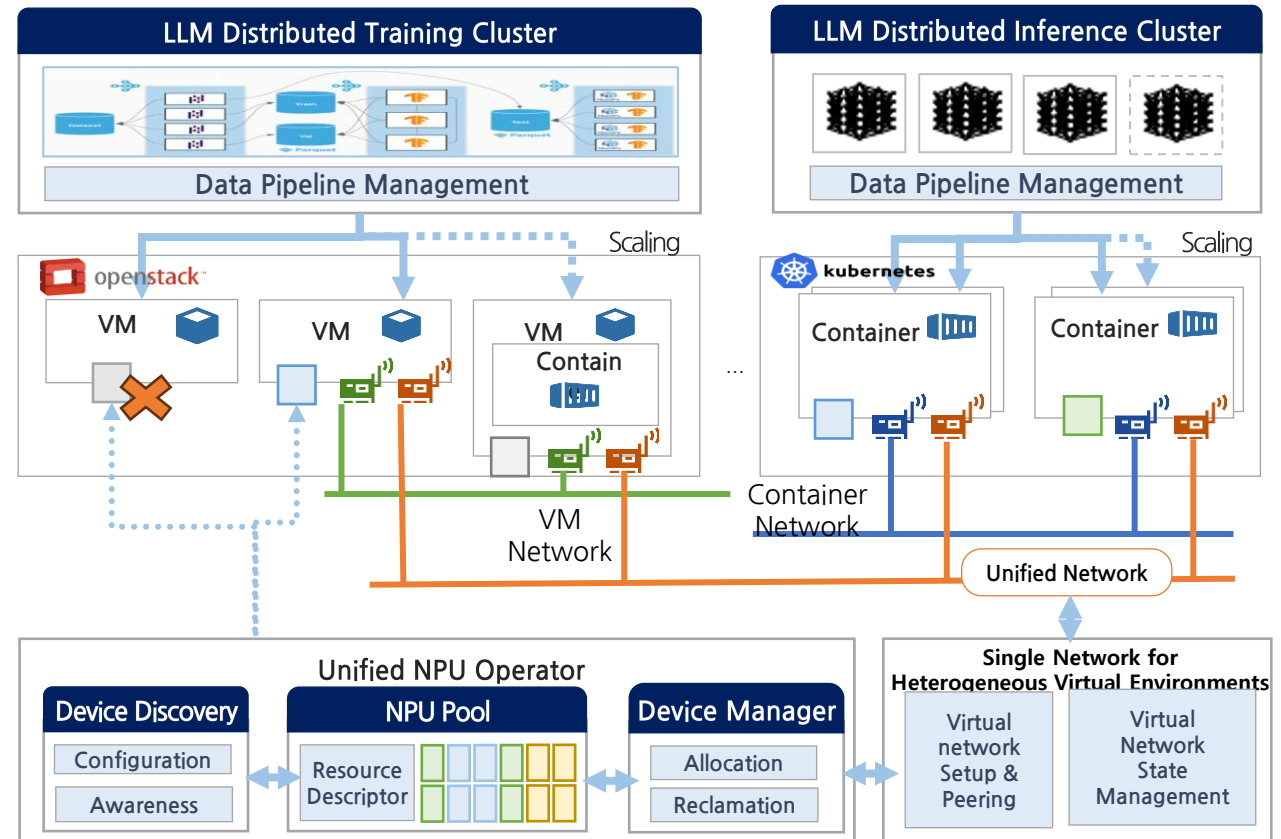




Heterogeneous Cloud Service Orchestration

Provisioning & Clustering of AI-Semiconductor Cloud Services

- Delivering large-scale training and inference on heterogeneous semiconductor clusters
- Seamless Scaling Without Service Interruption During Training & Inference
- Unified orchestration and portability across OpenStack and Kubernetes

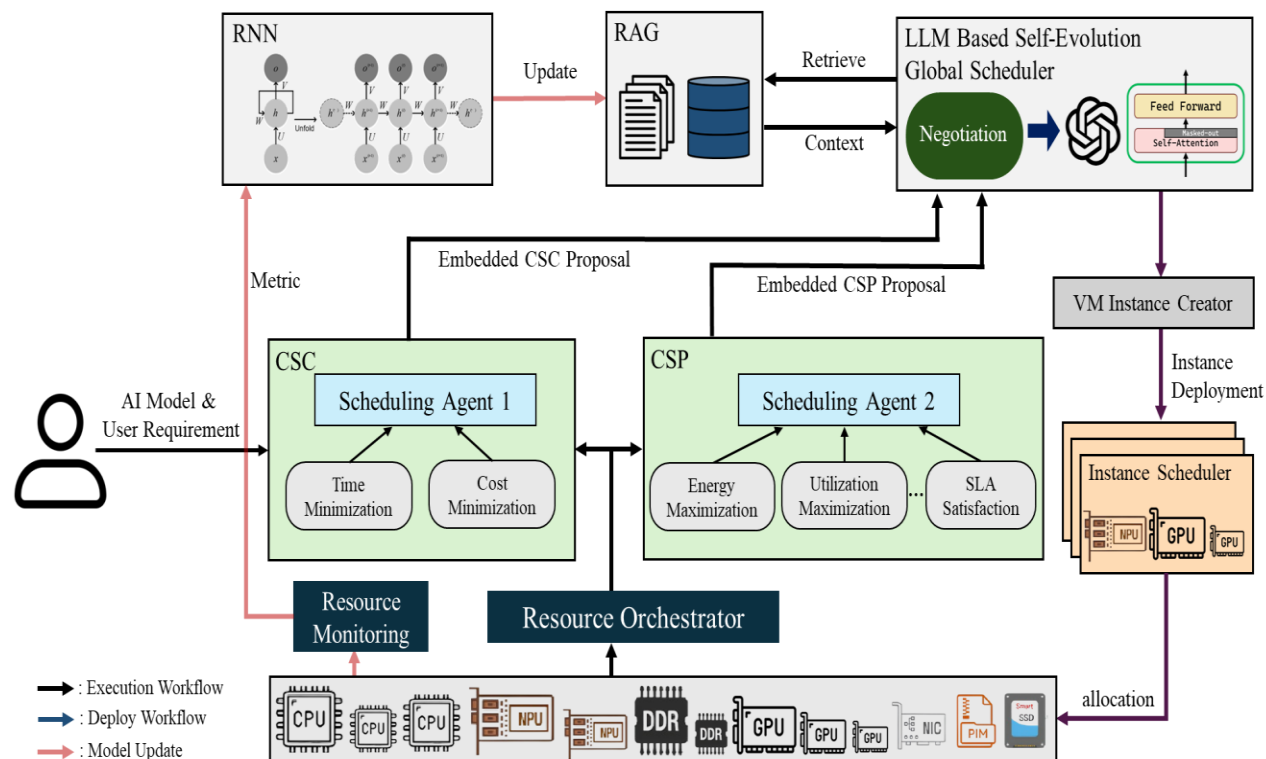




Heterogeneity-Aware Scheduling

Negotiation-Based Scheduling Customized to Multi-User Requirements

- Negotiation Algorithms Customized to User & Provider Requirements for LLM-Based Cloud Services
- Intelligent Scheduling for Heterogeneous AI-Semiconductor Clouds Driven by LLM Operation Characteristics

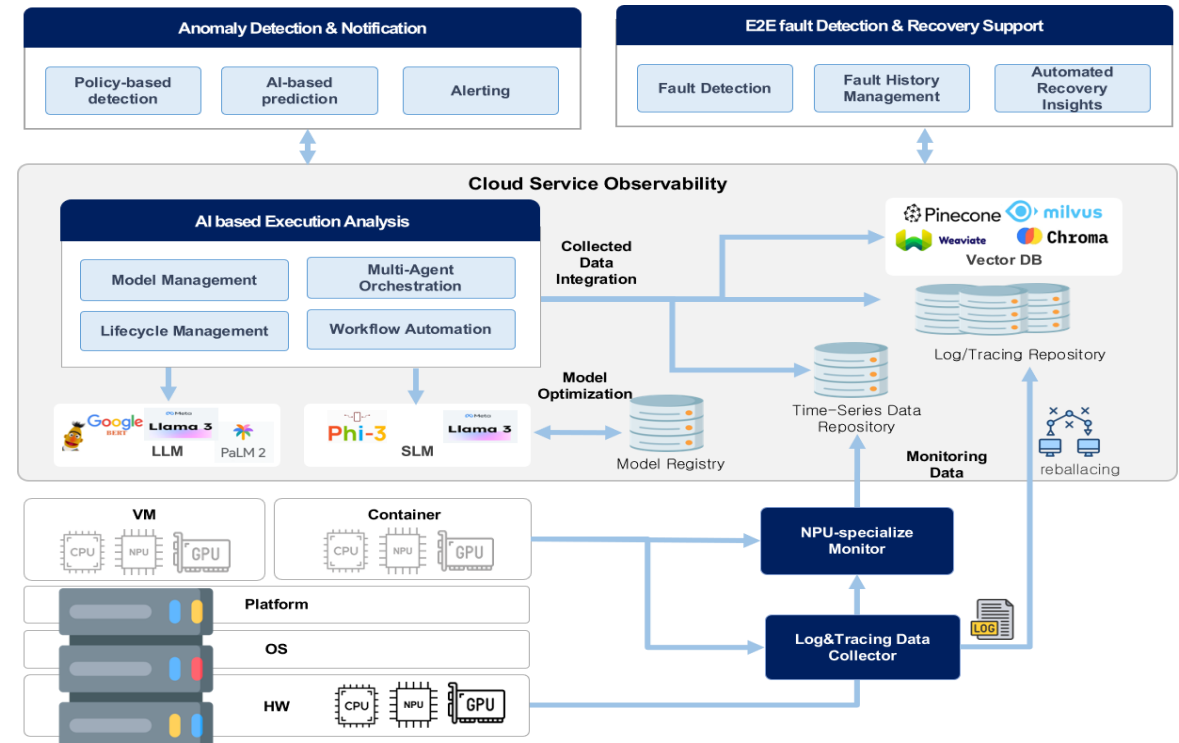




Cloud Service Observability

Full Observability across Cloud Service Layers

- Heterogeneous NPU-Specific Monitoring
- Cloud Service Log/Trace Data Collection
- Performance Degradation and Fault-symptom Detection with LLM-Based Predictive Models
- Proactive Anomaly Notification and Automated Fault Recovery

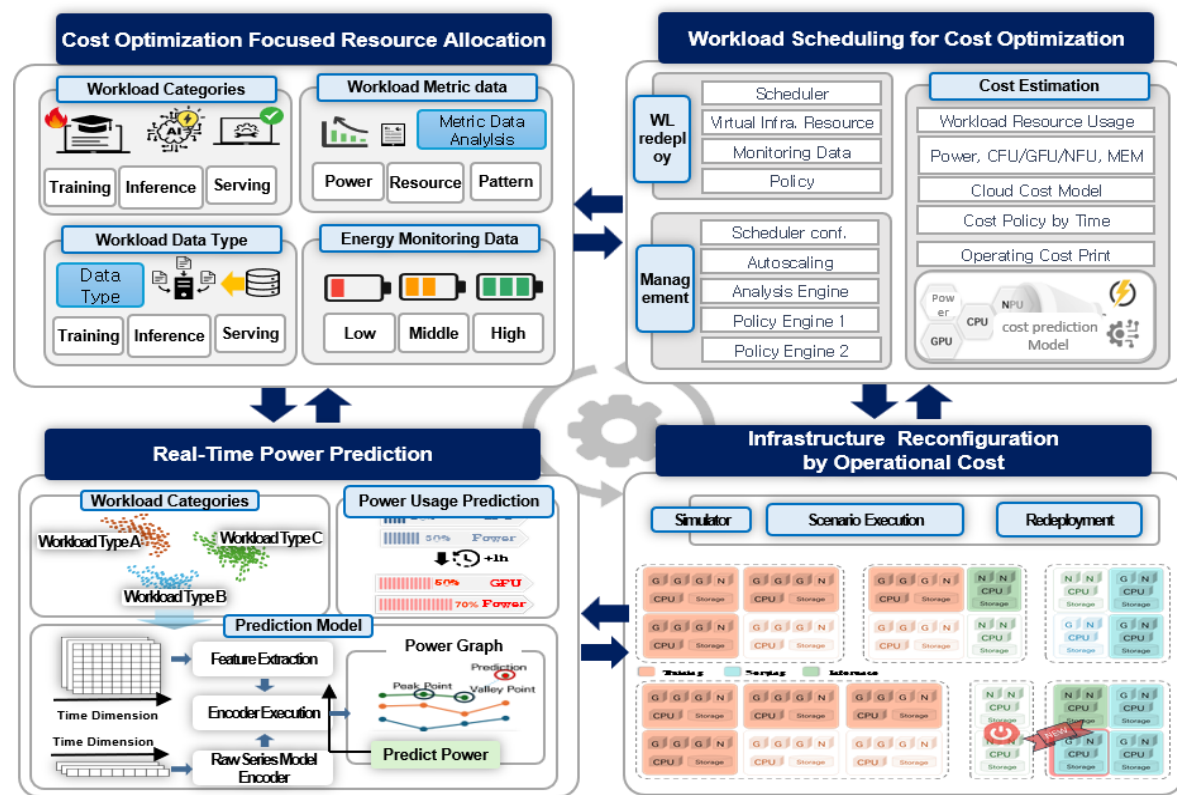




Opex Optimization for Cloud Services

Cost-Optimized Workload Allocation & Reconfiguration

- Fragmentation-Minimizing Workload Packing & Placement for Heterogeneous AI Semiconductors
- Cost-Saving Impact Modeling for Workload Performance vs. Operational Efficiency
- Resilience and efficiency by dynamically restructuring virtual cluster infrastructure and rebalancing

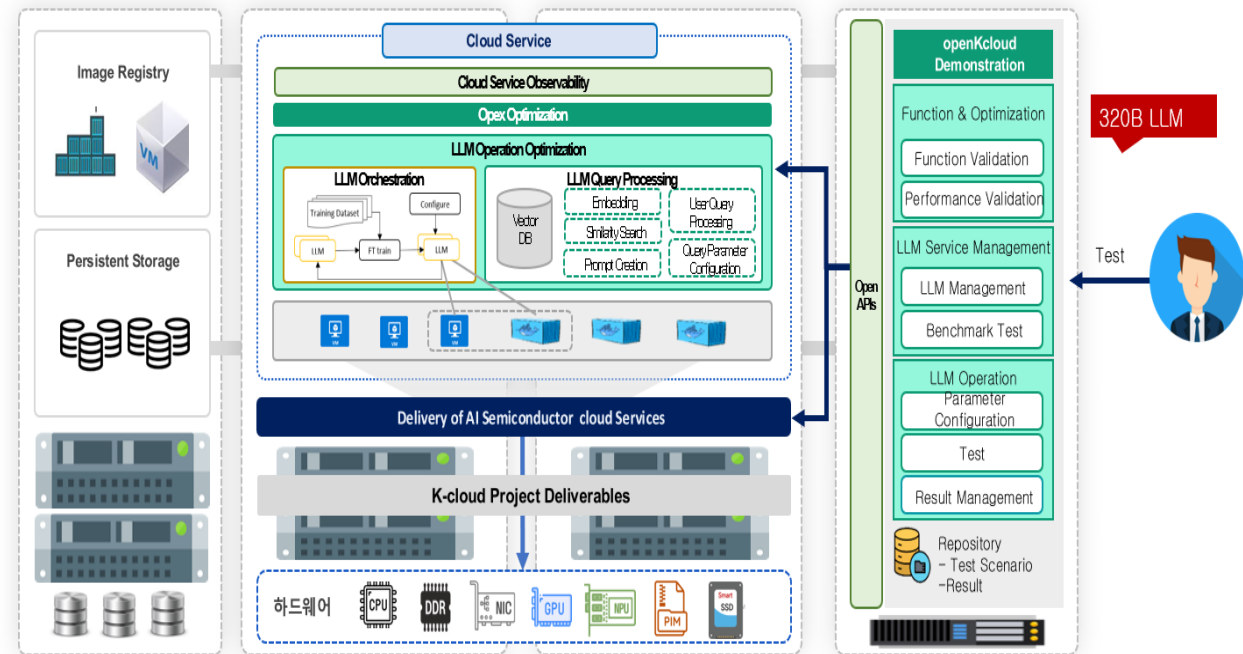




LLM Cloud Service Support

AI Semiconductor powered LLM Cloud Services

- Performance evaluation across representative, mainstream LLMs
- End-to-end tuning of retrieval, storage, and response latency
- Verifying functionality and measuring power-to-performance efficiency
- Full-stack cloud software validation based on Integrated K-Cloud Project Deliverables

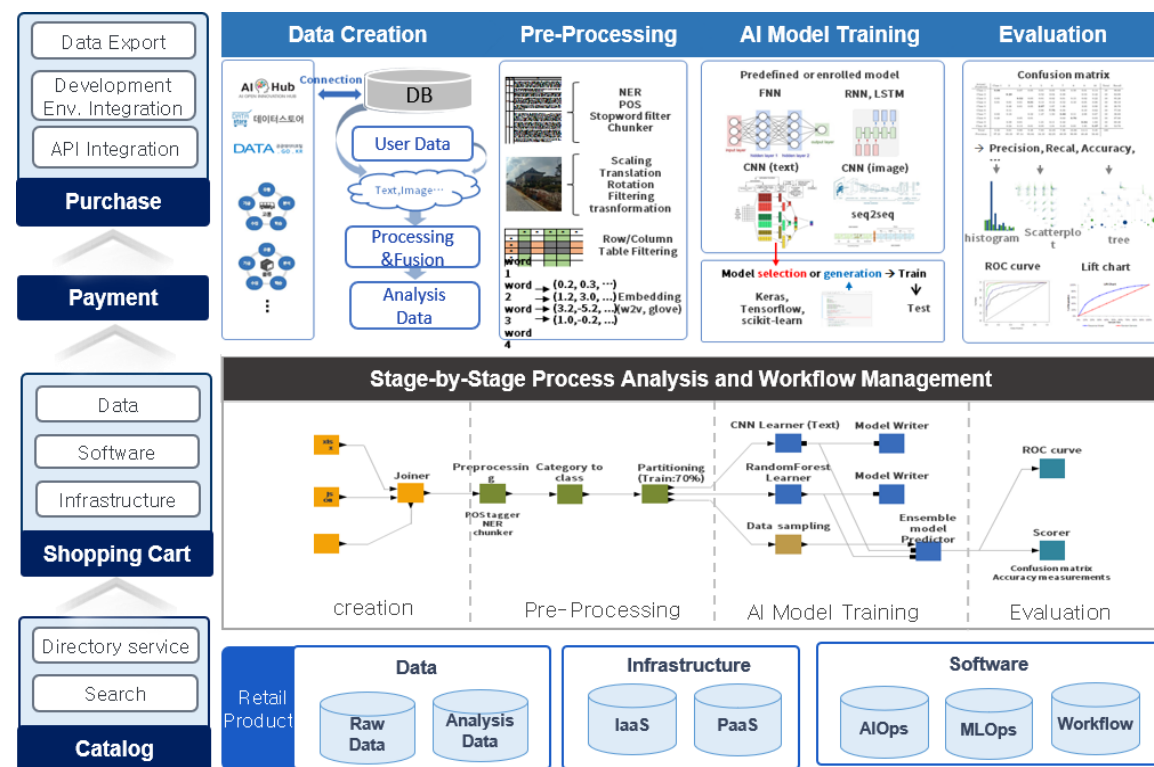




AI Semiconductor based Service Development

User-Friendly AI-Semiconductor Development Environment

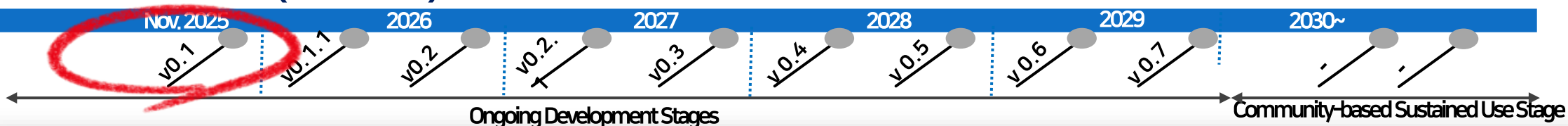
- Web-Based Cloud-Native IDE (VS Code-Like Experience)
- Implementation of Optimized Compilation Backends for AI Frameworks (e.g., PyTorch, TensorFlow)
- Marketplace for Inference & Training Services Powered by AI Semiconductors
- Secure Access with SSO and OAuth 2.0 Integration





Open Source Software Community

Biannual Release (Total 9th)



K-Cloud Community

OSS R&D Tech. Integration (Driving and Reflecting Requirements)

Biannual technical seminars

- (June, first half) AI Semiconductor Cloud Ecosystem
- (November, second half) OSS Developer Community
- (2025-08-07) Inter-Project Information Exchange Meeting

Heterogeneous Semiconductor Platform for MLOps

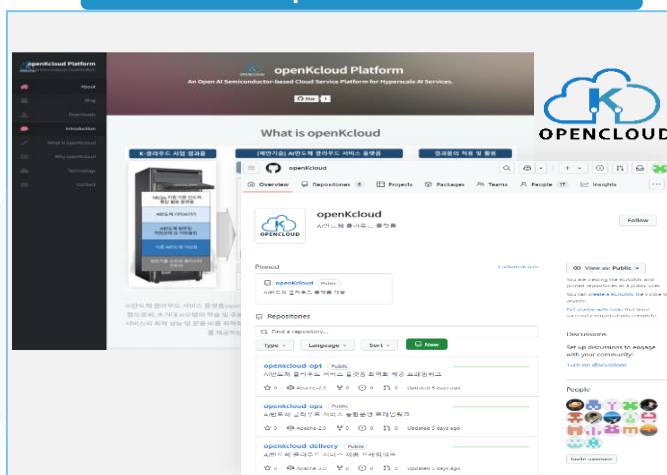
AI Semiconductor *Library*

Disaggregation & Pooling of AI Compute Resources

Virtualization for Heterogeneous Semiconductors

Composable Small-Scale Cluster Infrastructure

Open Source



Project Homepage: <https://openkcloud.github.io/>
Project Repository: <https://github.com/openkcloud>

Open Source Community

Domestic Open Source



Global Open Source



Talent Development



감사합니다.

